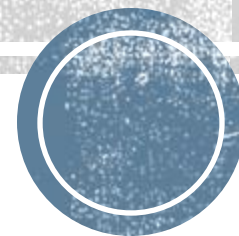


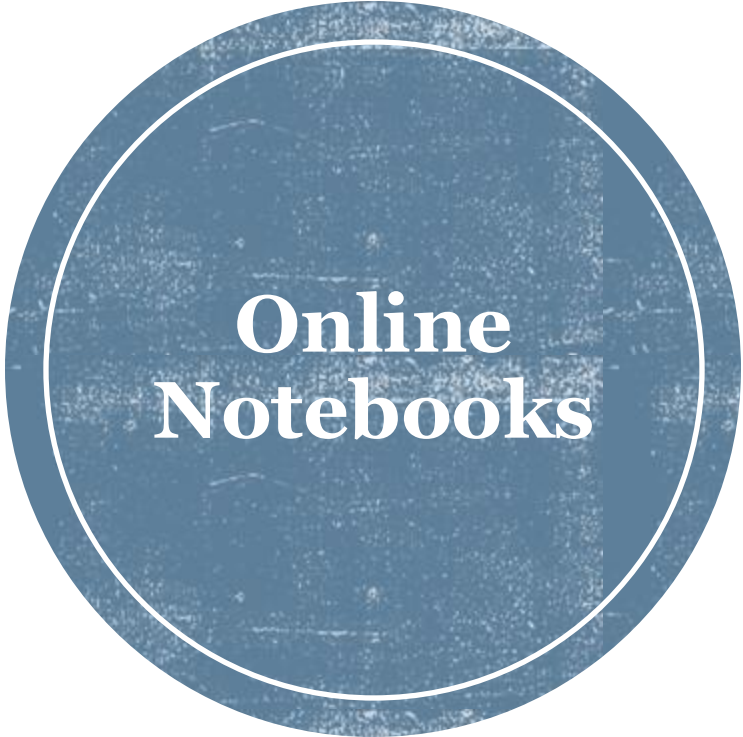
DSCoV Workshop, Oct 31, 2025



Uncovering Latent Structures in Data: An Introduction to Topic Modeling and Its Applications Beyond NLP

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Data Science Institute | Center for Computational Molecular Biology (CCMB)





Online Notebooks

- To follow the demos live, find the workshop's Google Colab notebooks here:

[github.com/xr-cc/
DSCoV-Workshop-on-Topic-Modeling](https://github.com/xr-cc/DSCoV-Workshop-on-Topic-Modeling)





If we have a collection of documents, can we tell what they are about?

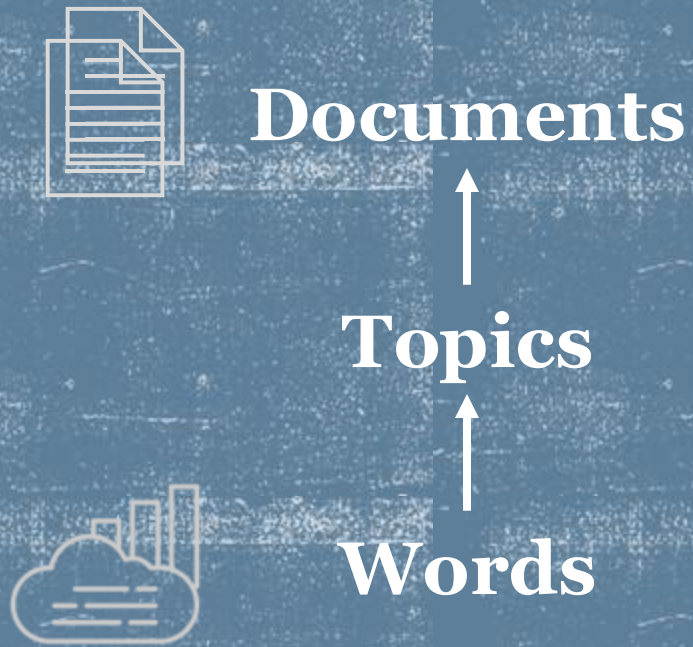
In other words, can we uncover the **hidden thematic structures** within them?

Topic Modeling



a collection of words that frequently appear together and represent some theme within the text





Top Words	Topic
market, stock, economy, inflation, trade	Economics
game, team, league, player, score	Sports
gene, mutation, expression, genome, DNA	Genomics
model, data, algorithm, prediction, training	ML

Formally,

- a **topic** is a **mixture of words** (i.e., a probability distribution over words), and
- each **document** is represented as a **mixture of topics**.

This is an **unsupervised learning** technique to extract hidden themes from text.



Think about how we read text.

Cancer survival in Germany
after the fall of the Iron Curtain

Two decades after the fall of the Wall, **cancer survival rates** in East and West Germany have become almost equal. This is reported in a recent study by a team of **epidemiologists** from the German **Cancer** Research Center and 11 German **cancer** registries.

Graphene plasmons go ballistic

Graphene combined with the
insulating **power** of boron nitride
enables **light** control in tiny
circuits with dramatically reduced
energy loss.



Documents



Where in the brain does creativity come from? Evidence from jazz musicians A new brain-imaging study out of Drexel University's Creativity Research Lab studied the brain activity of jazz guitarists during improvisation to show that creativity is, in fact, driven primarily by the right hemisphere in musicians who are comparatively inexperienced at improvisation. However, musicians who are highly experienced at improvisation rely primarily on their left hemisphere.

How your cozy fleece could be polluting the ocean Fleece is a wintertime staple but could be contributing to the next big ocean plastics problem: lint. The lightweight, cozy material sheds some of its synthetic microfibers each time it's washed, and this lint ultimately ends up in surface waters and in fish and other wildlife. An article in Chemical & Engineering News, the weekly news magazine of the American Chemical Society, looks into the problem and possible solutions.

Genetic information makes it safer to prescribe common blood thinner Doctors prescribing blood thinners have had to go through a lengthy trial-and-error process to arrive at the optimal dose for their patients. But now the process can be faster and safer, thanks to research conducted at Washington University School of Medicine in St. Louis.

MarrowQuant: A new digital pathology tool EPFL scientists have developed a digital pathology tool for quantifying bone marrow compartments in standard histological sections. Named 'MarrowQuant', the software makes it possible to examine bone marrow biopsies as well as to re-examine historical collections of bone-marrow samples and even old clinical trials.

Blood thinners may improve survival among hospitalized COVID-19 patients Research could change standard of care protocols to prevent clotting associated with coronavirus.

Rolling out an e-sticker revolution High-speed fabrication developed at KAIST can turn out adhesive and flexible electronic devices in any shape imaginable.

Moffitt researchers develop novel theoretical approach to reduce antibiotic resistance Moffitt Cancer Center researchers have developed a novel mathematical method inspired by Darwinian evolution to use current antibiotics to eliminate or reduce the development of antibiotic-resistant bacteria.

Huddling and a drop in metabolism allow penguins to survive the South Pole cold Emperor penguins endure their incubation and fast for four dark and bitterly cold months each year. The tight huddling among these South Pole penguins is a key energy-saving mechanism that allows them to endure the extremely harsh conditions. Huddling and a drop in metabolism allow penguins to survive the biting South Pole cold.

Donald Danforth Plant Science Center, MU to hire four joint researchers Officials from the University of Missouri and the Danforth Center announced a new collaborative agreement that will lead to the hiring of four new researchers who enhance plant science research to benefit the state, nation and world.

Graphene plasmons go ballistic: Graphene combined with the insulating power of boron nitride enables light control in tiny circuits with dramatically reduced energy loss.

Argonne to advance high performance computing in manufacturing Argonne awarded funding to partner with industry to advance the use of high performance computing in manufacturing.

New book examines states' influence on national voter turnout In her new book, 'How the States Shaped the Nation: American Electoral Institutions and Voter Turnout 1920-2000,' Melanie Jean Springer, UC Santa Cruz assistant professor of politics, examines voter turnout in the United States, one of the lowest in any developed democracy, and reform efforts to improve it.

Economic and social consequences of human mobility restrictions under COVID-19 A paper published in the Proceedings of the National Academy of Sciences by the research group coordinated by professor Fabio Panfili at Politecnico di Milano analyzes the steep fall on the Italian mobility network during the pandemic and reveals a counterintuitive and somewhat paradoxical result, since the contraction of mobility, in relative terms, has been more intense in the Regions where the diffusion of the virus has been negligible.

Special delivery: Nematode-infected insect cadavers A custom-made machine for packaging mealworms infected with beneficial nematodes could improve the delivery, timing and use of the wormlike organisms as biological control agents. The machine is the result of a cooperative research and development agreement involving US Department of Agriculture scientists and Southeastern Insectaries, Inc., of Perry, Ga.

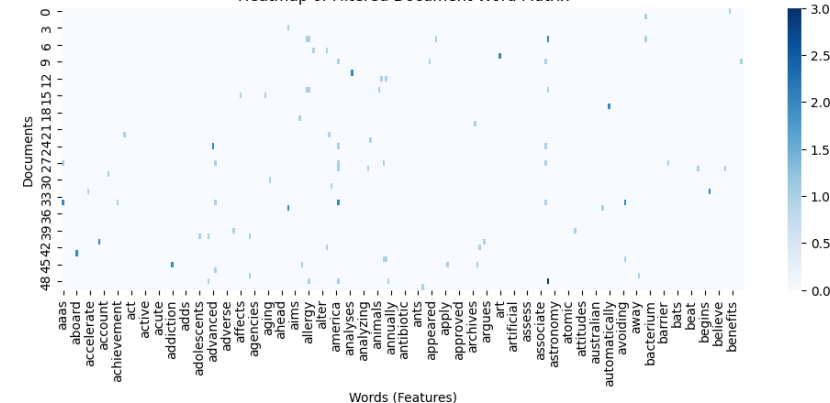
Severe sentences no deterrence, say criminologists Harsher sentences do not deter people from committing crimes, says a new report by University of Toronto criminologists.

Cancer survival in Germany after the fall of the Iron Curtain Two decades after the fall of the Wall, cancer survival rates in East and West Germany have become almost equal. This is reported in a recent study by a team of epidemiologists from the German Cancer Research Center and 11 German cancer registries.

Words



Heatmap of Filtered Document-Word Matrix



Count of each word
in each document



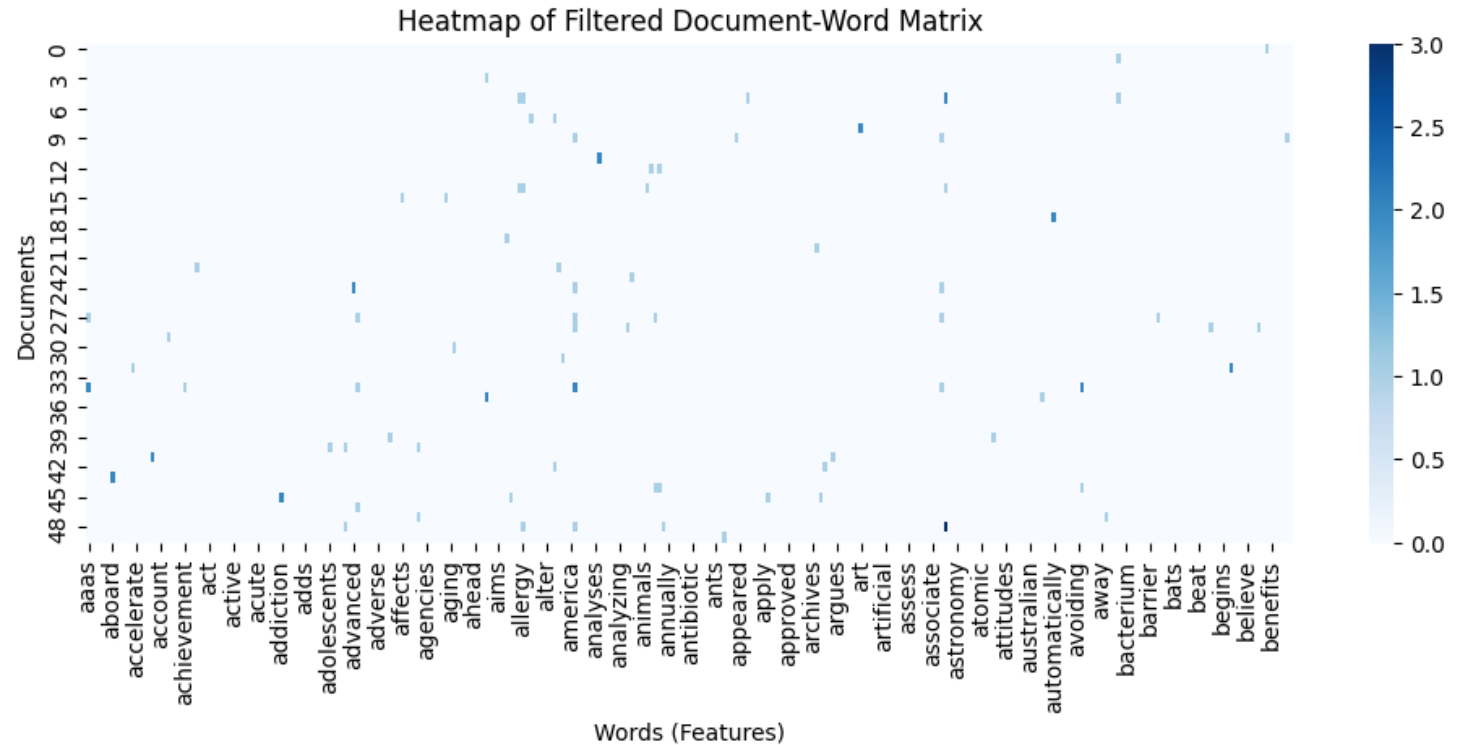
Suppose there are M documents and N words.

observed data

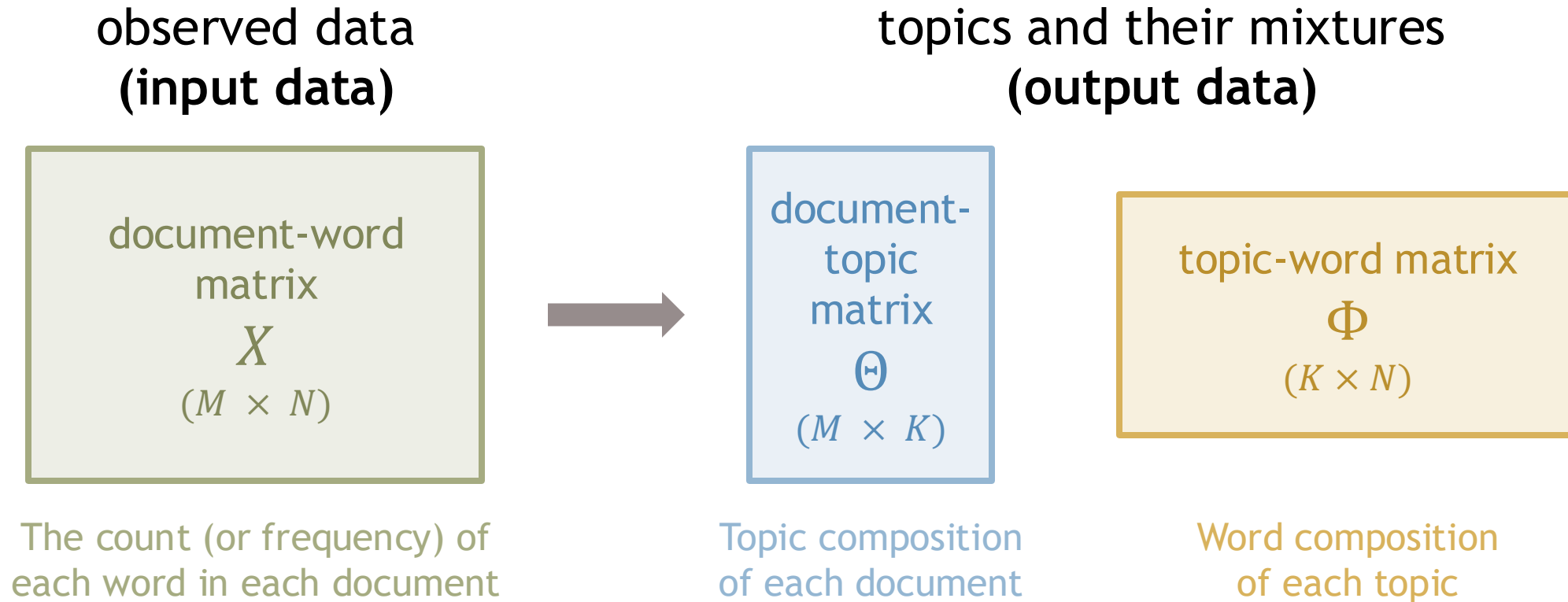
document-word
matrix

$$X$$
$$(M \times N)$$

The count (or frequency) of
each word in each document



Suppose there are M documents, N words, and K topics.



Topic Modeling Methods



Probabilistic / Generative Methods



Latent Dirichlet Allocation (LDA)



Matrix Factorization-Based Methods



Non-negative Matrix Factorization (NMF)



Deep Learning-Based Methods



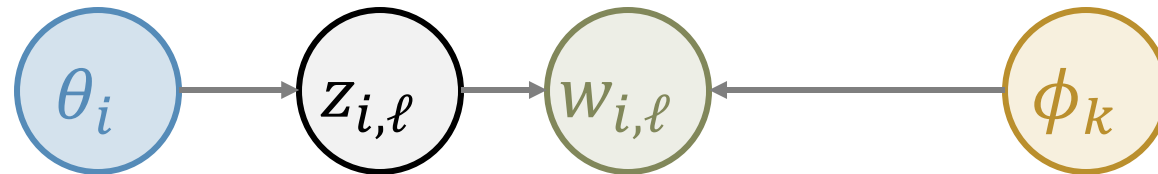
Specialized / Domain-Adapted Methods



Latent Dirichlet Allocation (LDA)

LDA treats X as being generated from hidden variables:

- Document i has a **topic distribution** θ_i .
- Topic k has a **word distribution** ϕ_k .
- Words are generated by sampling a topic from θ_i , and then a word from ϕ_k
- $z_{i,\ell}$ is the topic for the ℓ -th word in the i -th document: an integer in $\{1, \dots, K\}$
- $w_{i,\ell}$ is the ℓ -th word of the i -th document: an integer in $\{1, \dots, N\}$



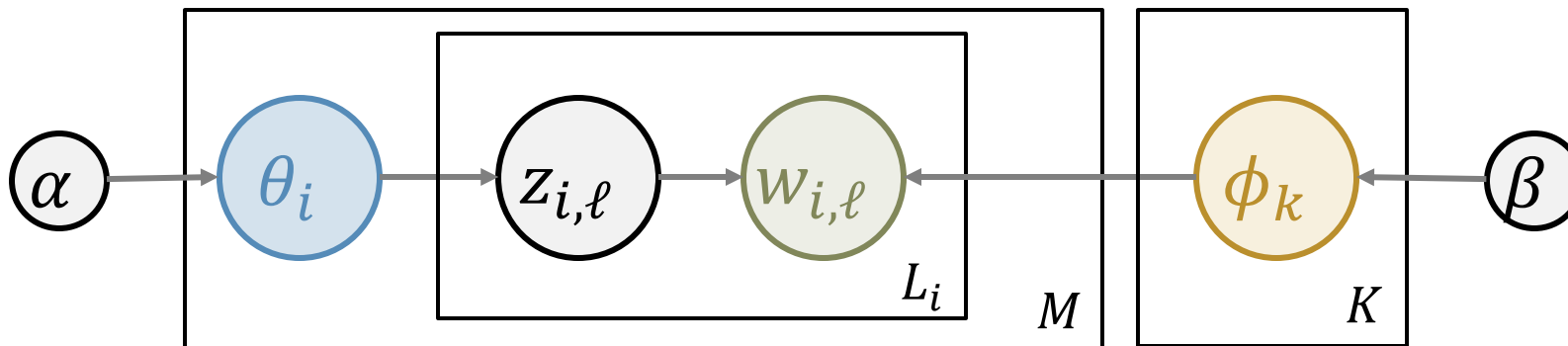
Note: L_i is the total number of words in document i ; $L_i \neq N$ (vocabulary size).

$$\sum_{k=1}^K \Theta_{ik} = 1 \text{ for } i = 1, \dots, M$$

$$\sum_{j=1}^N \Phi_{kj} = 1 \text{ for } k = 1, \dots, K$$

θ_i 's and ϕ_k 's are probability vectors.

- α and β are the parameters of the prior distribution over θ and ϕ , respectively: Dirichlet distributions: $\theta_i \sim \text{Dir}(\alpha)$, $\phi_k \sim \text{Dir}(\beta)$



Dirichlet distribution is a distribution over probability distributions over K elements.

- Distribution of order $K \geq 2$, parameterized by $\alpha_1, \dots, \alpha_K \geq 0$:

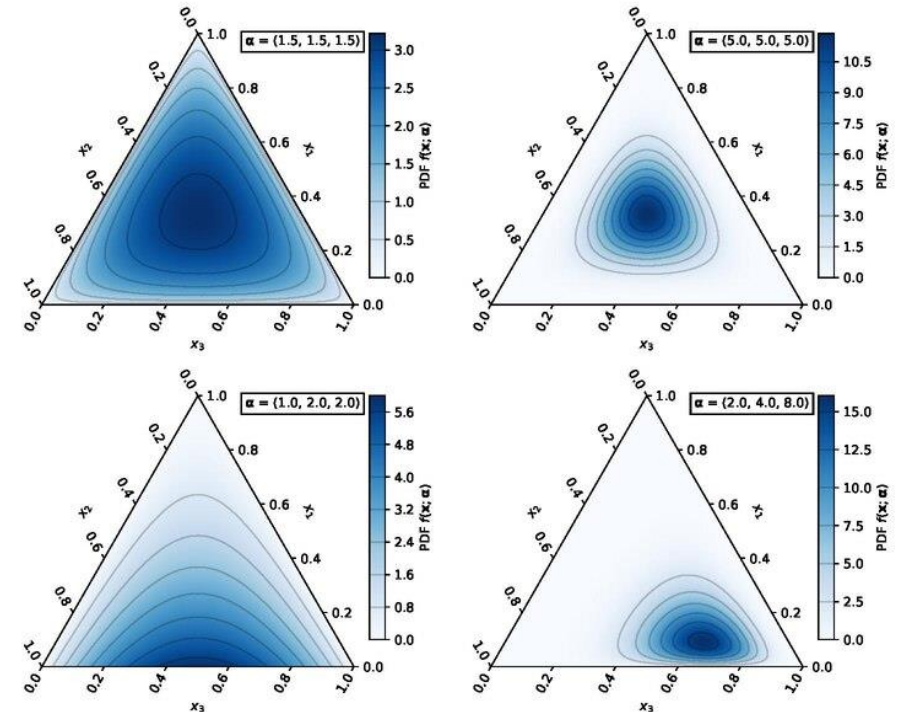
$$x \sim \text{Dir}(\alpha), \quad 0 \leq x_1, \dots, x_K \leq 1, \quad \sum_k x_k = 1$$

- Probability density function:

$$f(x_1, \dots, x_K; \alpha_1, \dots, \alpha_K) = \frac{1}{B(\alpha)} \prod_{i=1}^K x_i^{\alpha_i - 1}$$

- $B(\alpha)$ is the multivariate beta function: $B(\alpha) = \frac{\prod_{i=1}^K \Gamma(\alpha_i)}{\Gamma(\alpha_0)}$ with $\alpha_0 = \sum_{i=1}^K \alpha_i$ and the gamma function $\Gamma(\cdot)$

- Mean: $E[X_i] = \frac{\alpha_i}{\alpha_0}$ Variance: $\text{Var}[X_i] = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)}$
- It is the conjugate prior distribution of the multinomial distribution.



Wikipedia



Generative Process of LDA

For each topic $k \in \{1, \dots, K\}$:

$$\phi_k \sim \text{Dir}(\beta)$$

Draw its distribution over words

For each document $i \in \{1, \dots, M\}$:

$$\theta_i \sim \text{Dir}(\alpha)$$

Draw its distribution over topics

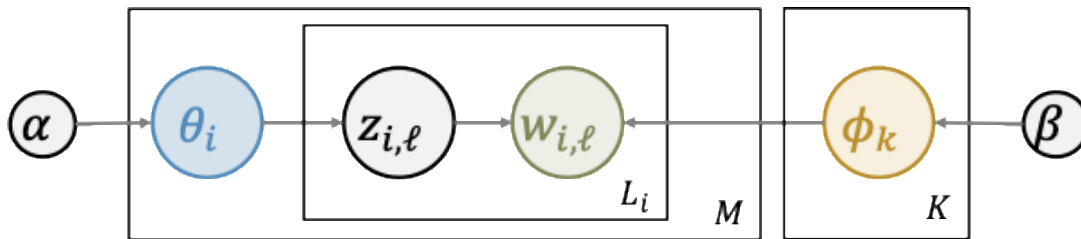
For each word $\ell \in \{1, \dots, L_i\}$:

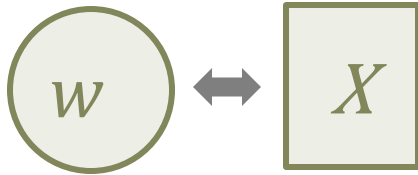
$$z_{i\ell} \sim \text{Mult}(1, \theta_i)$$

Draw the topic assignment → Each topic is defined as a Multinomial distribution over the vocabulary, parameterized by θ_i .

$$w_{i\ell} \sim \phi_{z_{i\ell}}$$

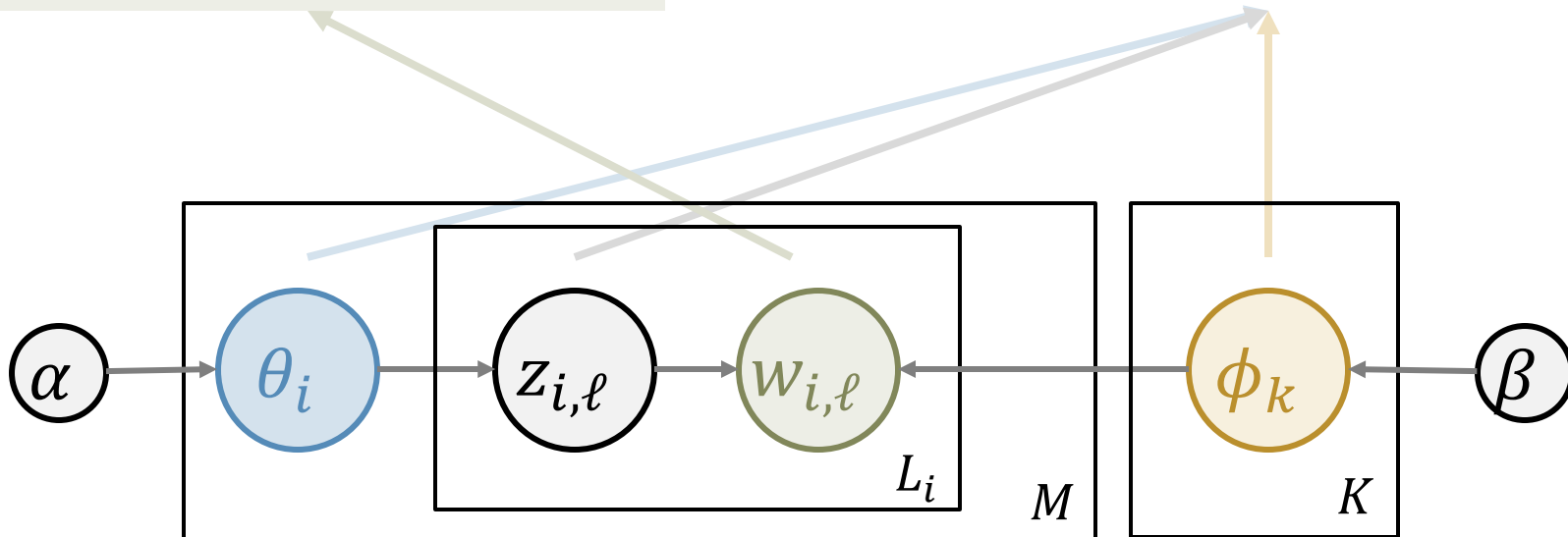
Draw the word





- The only **observed variables** are the words in the documents.

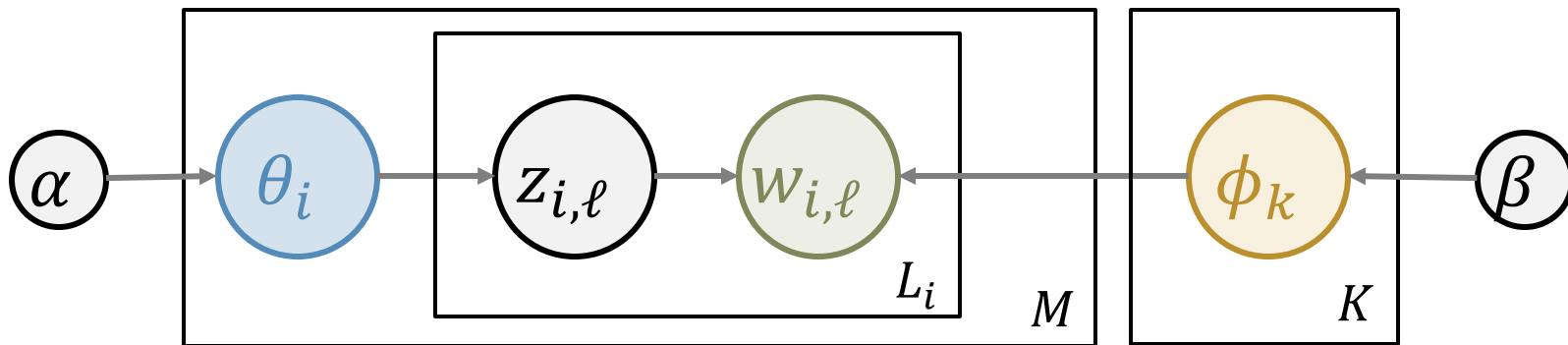
- The topic for each word, the **distribution over topics for each document**, and the **distribution of words per topic** are all **latent variables** in this model.



- Skipping the math here ...
- The joint probability is

$$p(w, z, \theta, \phi | \alpha, \beta) = \prod_k p(\phi_k | \beta) \prod_i \left[p(\theta_i | \alpha) \prod_{\ell} p(z_{i,\ell} | \theta_i) p(w_{i,\ell} | z_{i,\ell}, \phi) \right]$$

- **Goal of LDA:** find the parameters that maximize this probability.
- This is usually solved by MCMC or Variational EM (VEM).



TRENDING NEWS RELEASES



ESMO 2025: Trial results show belzutifan shrinks rare neuroendocrine tumors and improves symptoms in patients

UNIVERSITY OF TEXAS M. D. ANDERSON
CANCER CENTER



Important phenomenon discovered in the Arctic – could boost marine life

UNIVERSITY OF COPENHAGEN



Microscopic DNA 'flowers' could deliver medicine exactly where it's needed

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

New white paper urges policymakers to modify practice laws to unlock AI's full potential in health

WEST HEALTH INSTITUTE

Data for NLP Demo

- Press releases from the **EurekAlert!** Platform collected by Zhang et al. 2025.
- **EurekAlert!** is an online science news service featuring health, medicine, science and technology news from leading research institutions and universities, sponsored by American Association for the Advancement of Science (AAAS).

[1] Zhang, Jingwen; Jonathan Dudek; Alysson Mazoni; Enrique Orduna-Malea; Rodrigo Costas, 2025, "EurekAlert!: An open dataset for science communication research", <https://doi.org/10.34894/EZO4JE>, DataverseNL, V2



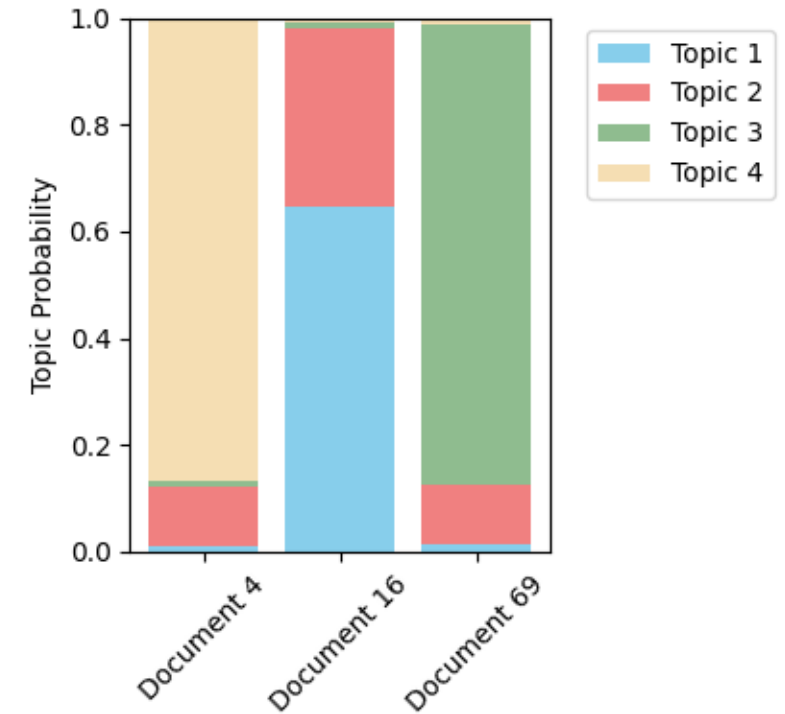


Notebook Demo

- Implementation in Python:
 - `LatentDirichletAllocation` from `sklearn.decomposition` ✓
 - `LdaModel` from `genism.models` (`genism` is a Python library for topic modelling, document retrieval by similarity, and other NLP tasks.)
- Implementation in R:
 - multiple packages, e.g., `topicmodels` (<https://www.rdocumentation.org/packages/topicmodels>)

Toy Example LDA Results

Topic	Top words
Topic 1	cancer patients disease cells risk treatment heart drug blood cell
Topic 2	brain children change human climate health species likely women years
Topic 3	energy light nasa way technology water earth high carbon tropical
Topic 4	health million care professor american society award grant medical center



Document 4: Ratification of human rights treaties makes no difference to health status A paper in this week's edition of the Lancet concludes that whether or not a country has ratified UN human rights treaties has no effect on the health status ...

Document 16: A step forward in the struggle against neuroblastoma, a type of childhood cancer A protein called CD44 makes it possible to identify the population of mother cells that are responsible for the aggressive nature and low survival rate ...

Document 69: UNH scientist: Cosmic rays threaten future deep-space astronaut missions Crewed missions to Mars remain an essential goal for NASA, but scientists are only now beginning to understand and characterize the radiation hazards ...



Non-negative Matrix Factorization (NNMF or NMF)

- Approximate X as the product of two smaller non-negative matrices.

$$X \approx WH, \quad W \in \mathbb{R}_+^{M \times K}, H \in \mathbb{R}_+^{K \times N}, \quad X_{i,j} \approx \sum_{k=1}^K W_{i,k} H_{k,j}$$

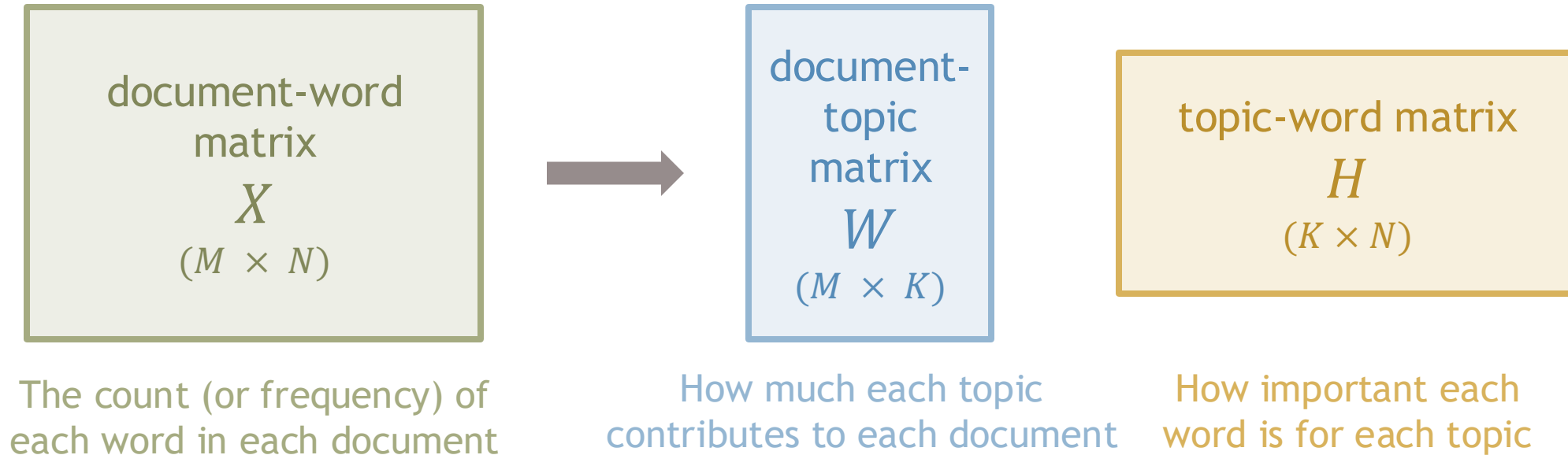
- This is done by solving

$$\min_{W, H \geq 0} \|X - WH\|_F^2 \quad \text{or} \quad \min_{W, H \geq 0} D_{\text{KL}}(X \| WH)$$

subject to non-negativity constraints.

- Goal of NMF:** minimizes reconstruction error under nonnegativity.





Note: W and H can be normalized to obtain analogs of Θ and Φ .





Notebook Demo

- Implementation in Python:
 - NMF from `sklearn.decomposition` 
- Implementation in R:
 - multiple packages, e.g., NMF (<https://github.com/renozao/NMF>) 

Toy Example NMF Results

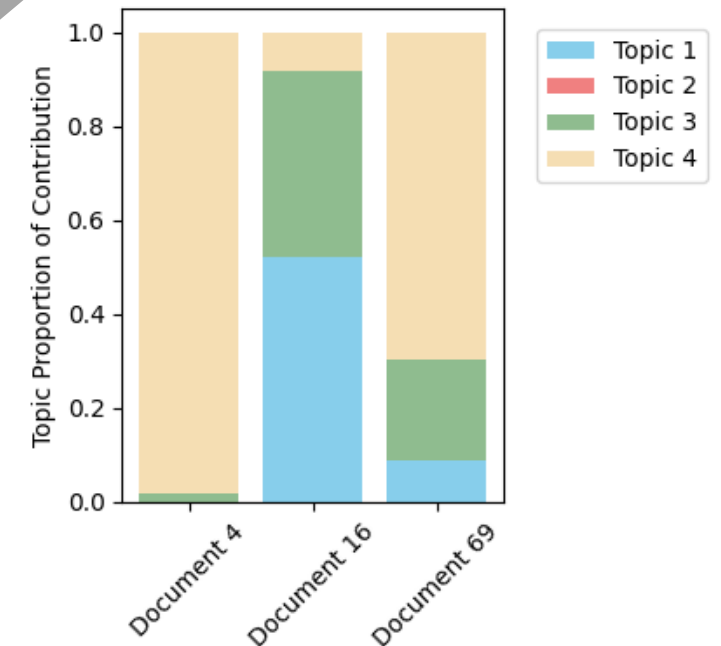
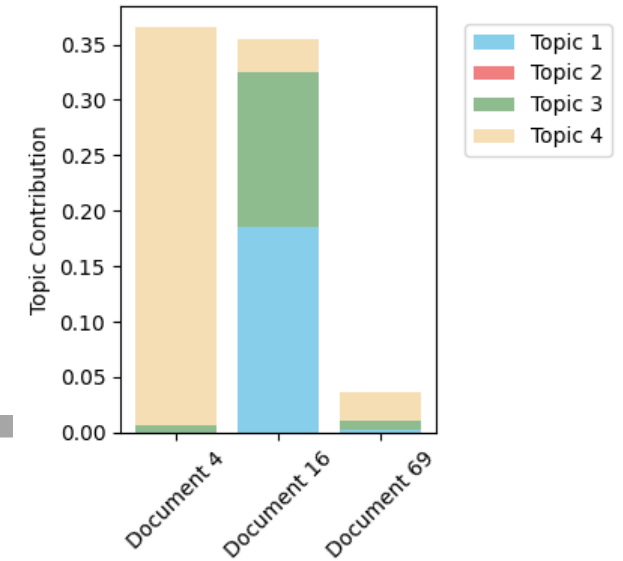
Topic	Top words
Topic 1	cancer, breast, center, patients, treatment, prostate, women, lung, therapy, drug
Topic 2	disease, patients, risk, heart, blood, treatment, women, high, associated, alzheimer
Topic 3	cells, cell, brain, stem, human, protein, immune, blood, gene, discovered
Topic 4	health, children, care, million, school, medicine, public, medical, center, grant

Document 4: Ratification of human rights treaties makes no difference to health status A paper in this week's edition ...

Document 16: A step forward in the struggle against neuroblastoma, a type of childhood cancer A protein ...

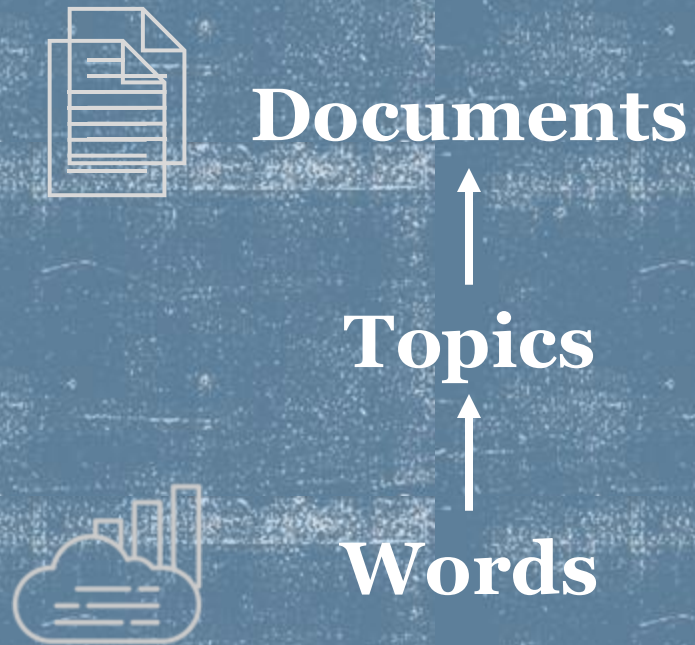
Document 69: UNH scientist: Cosmic rays threaten future deep-space astronaut missions Crewed missions to Mars ...

Normalization



Concept	LDA	NMF
Framework	Probabilistic (generative model)	Matrix decomposition
Inputs	Word counts	Word counts or TF-IDF
Outputs	Topic-word & document-topic distributions	Topic-word & document-topic weight matrices
Constraints	Simplex (probabilities sum to 1)	Non-negativity (no sum-to-one constraint)
Inference	Bayesian / variational inference	Gradient-based / iterative optimization

This is an **unsupervised learning** technique
to extract ~~hidden themes~~ from ~~text~~.
latent structures data



	Single-Cell Transcriptomics	Cancer Genomics	User Profiling
Entities	Cells	Tumor samples	Users
Latent Groups	Cell types	Mutational signatures	Interests
Features	Genes	Mutation types	Activities

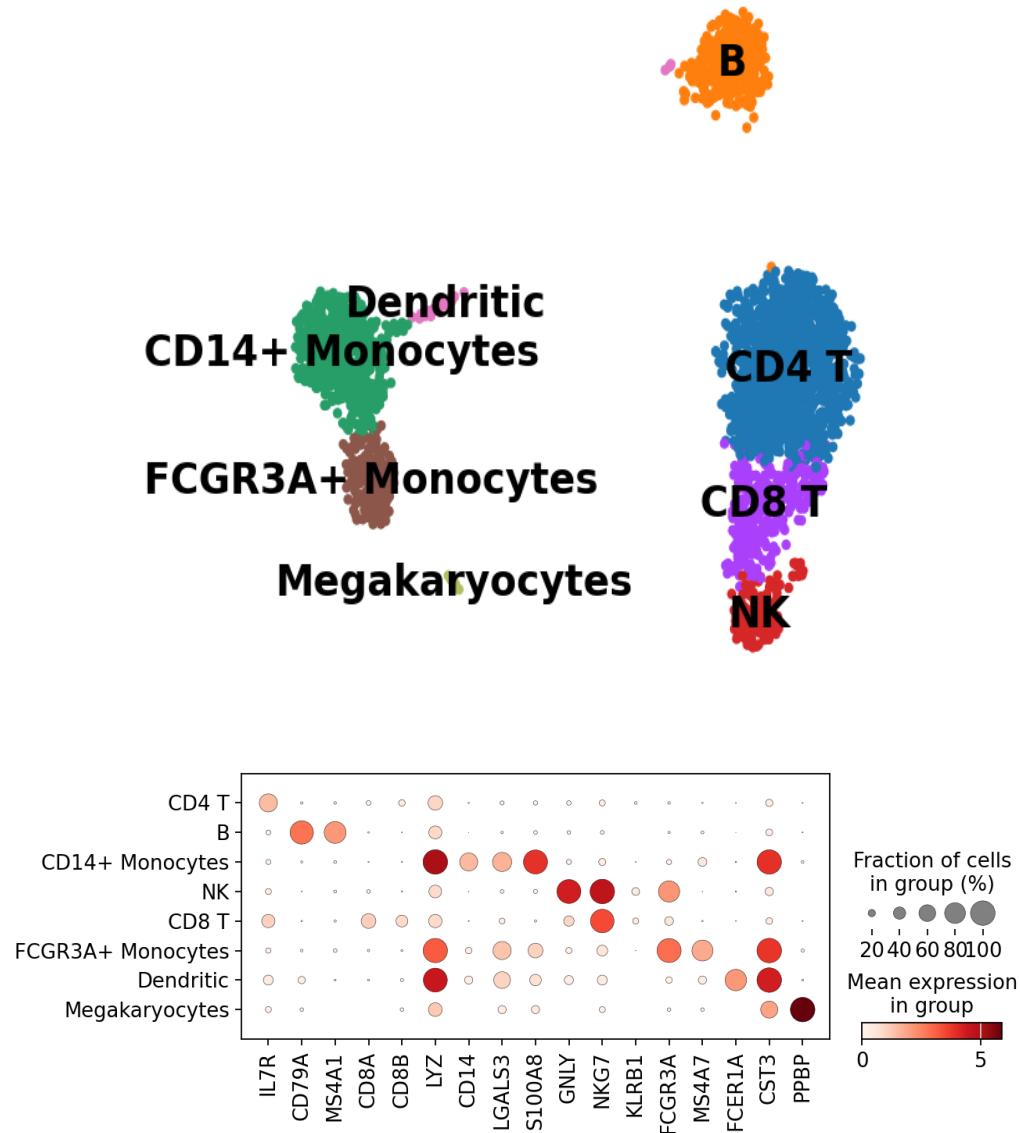


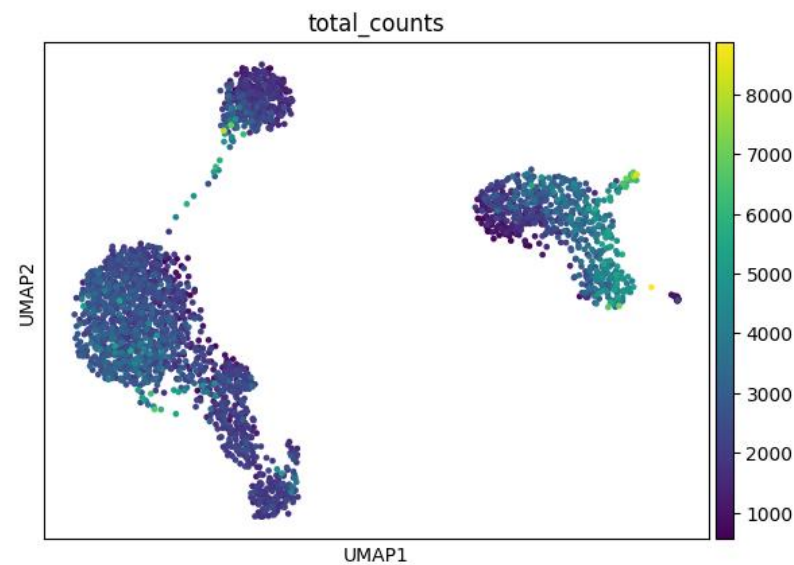
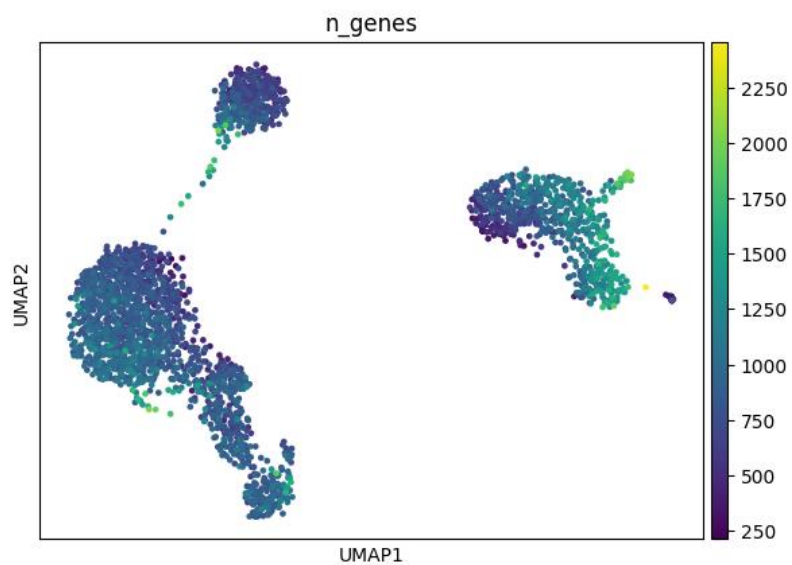
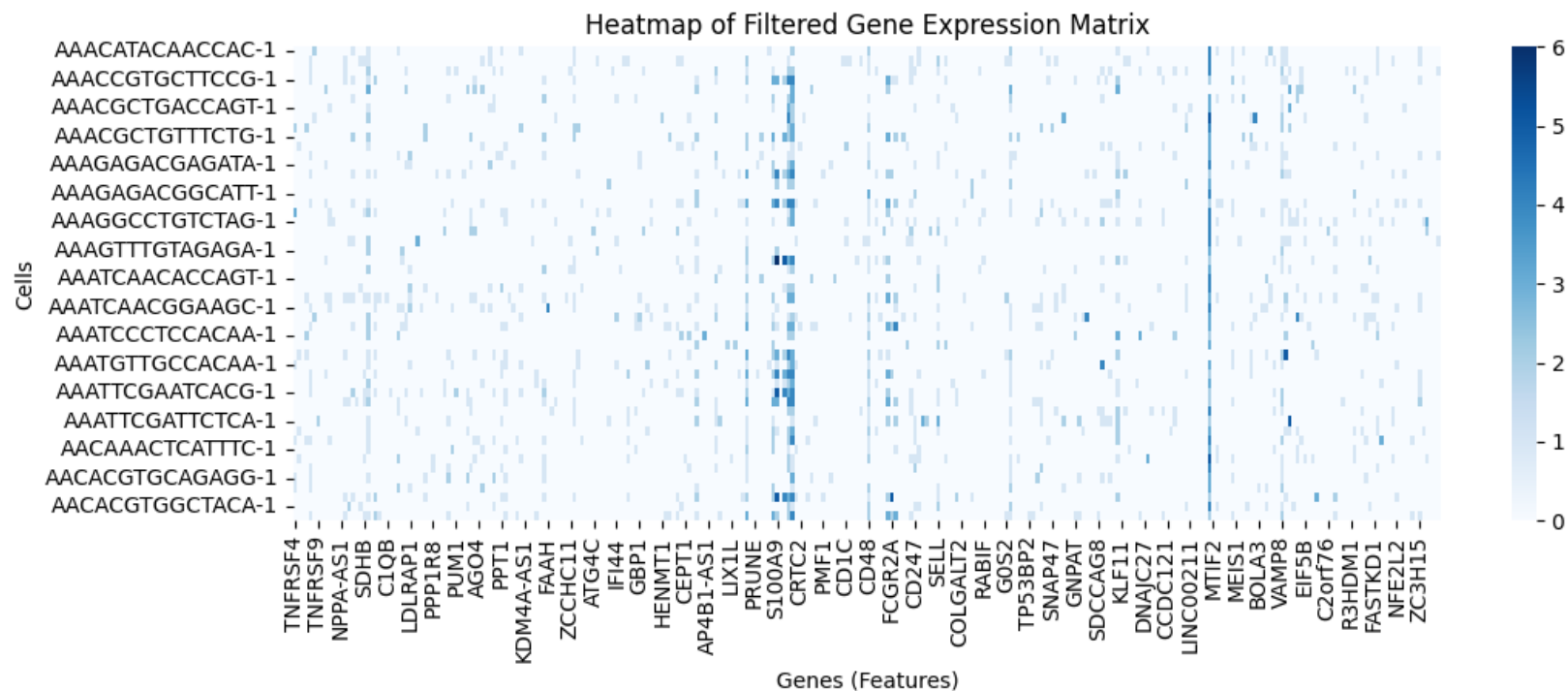
Data for Single-Cell Transcriptomics Demo

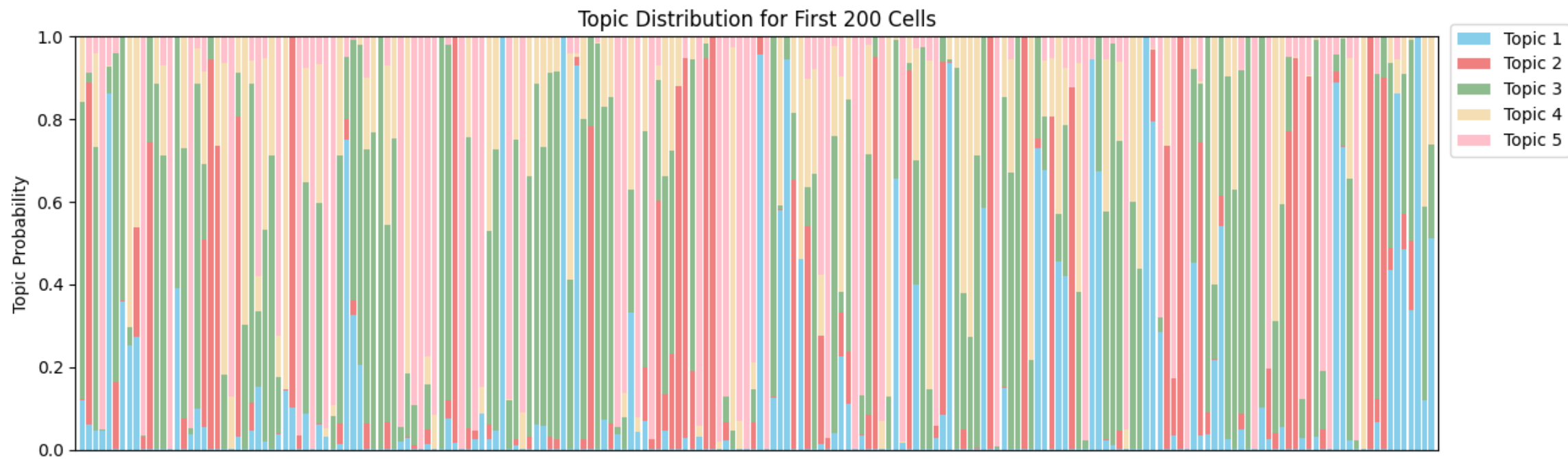
- Single-cell RNA sequencing (scRNA-seq) data of human peripheral blood mononuclear cells (PBMCs) of a healthy donor from 10x Genomics.
- Each value in the data matrix (M cells $\times$ N genes) is the gene expression count for a gene in a cell.

[2] 10x Genomics. 2016. *3k PBMCs from a healthy donor (v1.1.0)*. 10x Genomics Datasets.

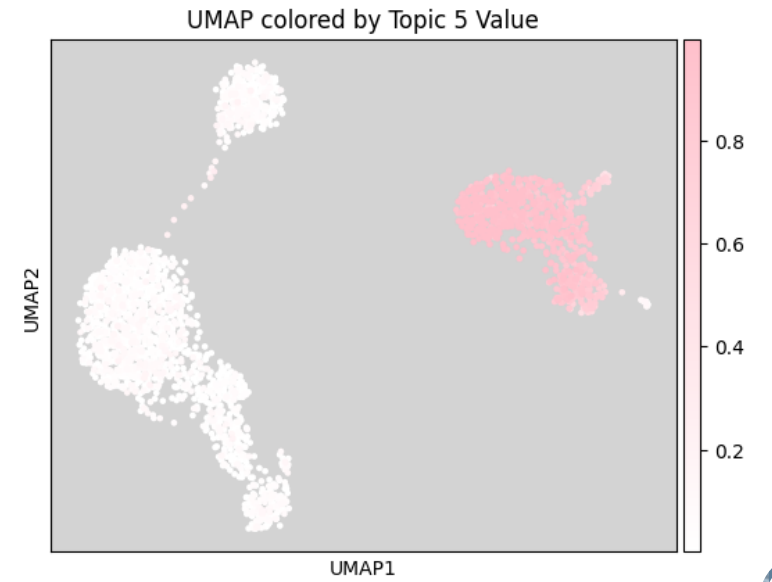
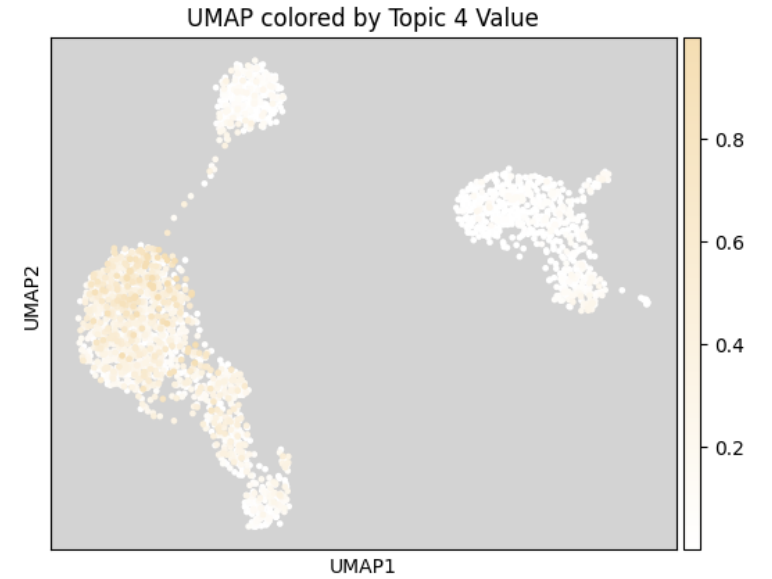
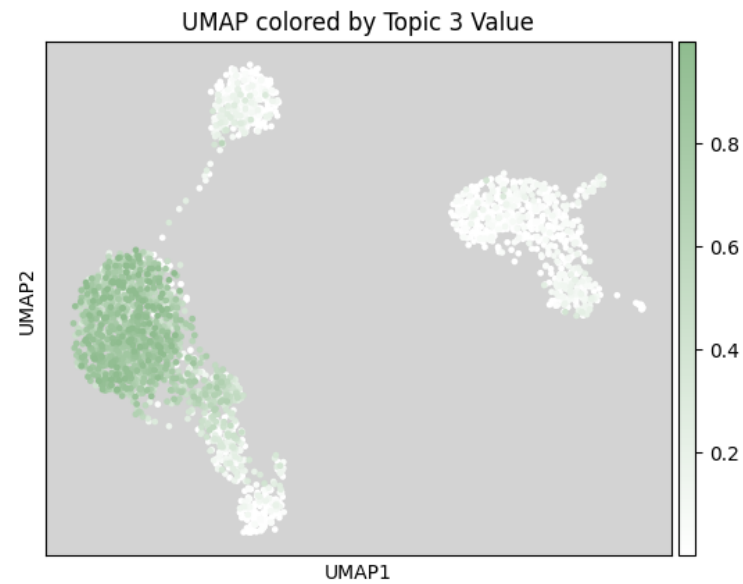
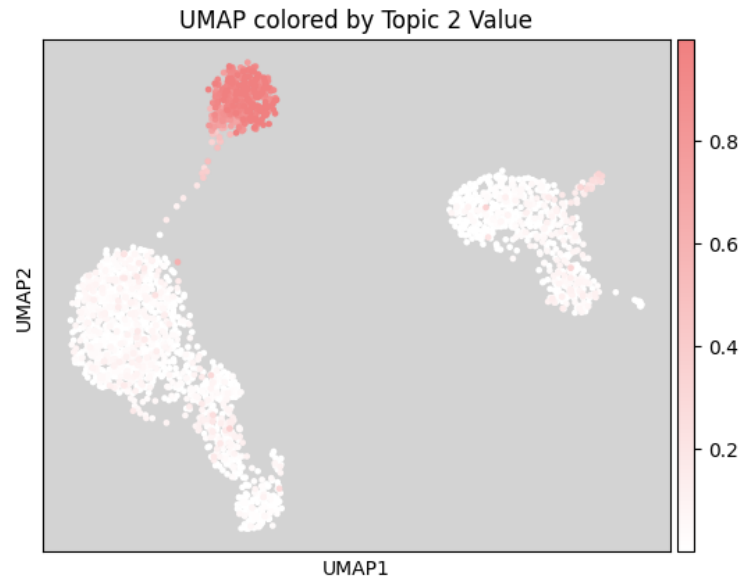
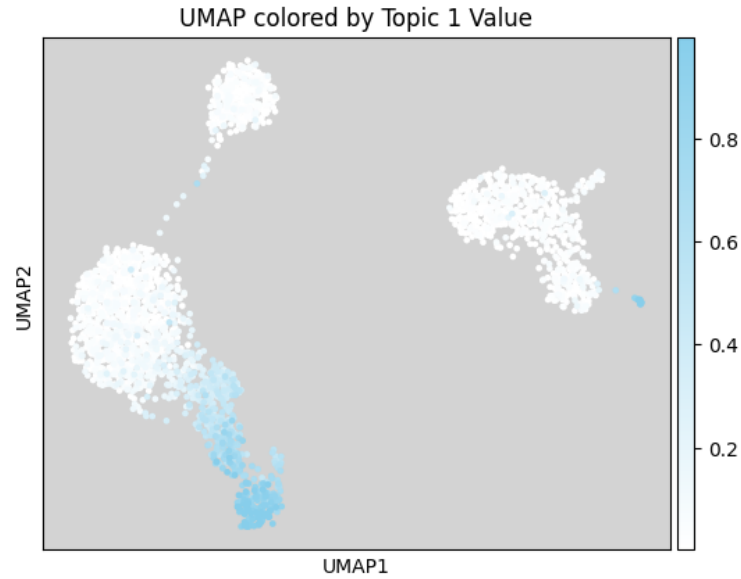
<https://support.10xgenomics.com/single-cell-gene-expression/datasets/1.1.0/pbmc3k>



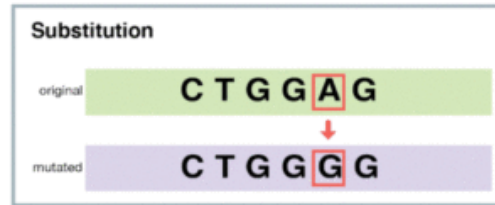




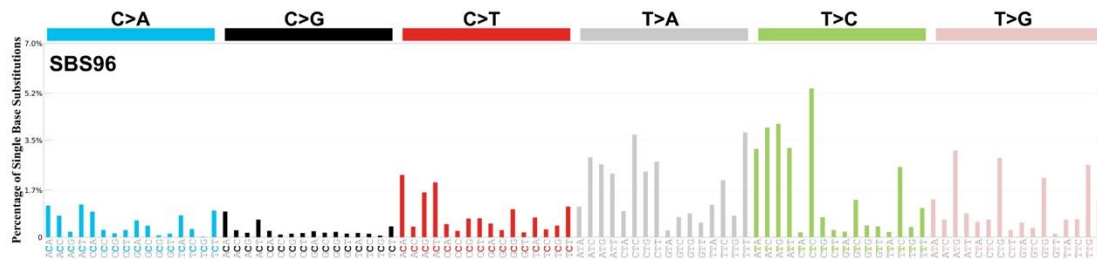
scRNA-seq LDA Results (K=5)



A Demo in R



<https://evolution.berkeley.edu/dna-and-mutations/types-of-mutations/>



<https://cancer.sanger.ac.uk/signatures/sbs/sbs96/>

Data for Mutational Signature Analysis Demo

- SBS96 mutation count matrix of 50 samples from the TCGA bladder cancer dataset.
- Each value in the data matrix (M samples $\times$ $N = 96$ mutation types) is the count for a mutation type in a sample.

[3] Example mutation count matrix were loaded from `mutSignatures::mutSigData` (Fantini et al., 2020). Data comes from bladder cancer genomes from TCGA.



Interpreting Results

- How to interpret each topic?
- How to choose K ?
- How to evaluate the performance of topic modeling?



Interpreting Results

- How to interpret each topic?

A **topic** is usually summarized by its high probability words.

Top Words	Topic
market, stock, economy, inflation, trade	Economics
game, team, league, player, score	Sports
gene, mutation, expression, genome, DNA	Genomics
model, data, algorithm, prediction, training	ML



Interpreting Results

- How to choose K ?

TL;DR

Pick a range.

To get a single K ,

- Perform topic modeling with each K a few times.
- Evaluate performance of each K .
- Choose K at the elbow.



Interpreting Results

- How to evaluate the performance of topic modeling?
 - Manually inspect top words
 - Compute metrics, e.g., **Coherence score** measures the semantic similarity between the top-n words in a topic → higher is usually better
 - Evaluate in downstream applications



Summary

- Introduces **topic modeling**, a family of unsupervised machine learning techniques for uncovering latent themes and structures in large text or other datasets.
- Covers common approaches such as **Latent Dirichlet Allocation (LDA)** and **Non-negative Matrix Factorization (NMF)**.
- Demonstrates its **interdisciplinary applications** (in Python & R) on **NLP** and **biological** data.





GitHub

- The workshop materials are available online at <https://github.com/xr-cc/DSCoV-Workshop-on-Topic-Modeling>.
- If you have any questions or would like to discuss potential applications of this technique to your own project/data, I'd be happy to chat.



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Data Science Institute



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Molecular Biology

